

APPENDIX: COHOMOLOGY WITH PROPER SUPPORT AND THE CONSTRUCTION OF THE FUNCTOR $f^!$

PIERRE DELIGNE

This is a completed version of a letter from P. Deligne to R. Hartshorne (letter of March 3, 1966). The footnotes were added by the copyist.

Verdier showed that, in the topological case, the formalism of Poincaré duality reduces to local questions upstairs (see [1]). In order to transpose his construction to the schematic setting, one needs a theory of cohomology “with proper support” for coherent sheaves.

Unless explicitly stated otherwise, all preschemes considered below are Noetherian, and all sheaves are quasi-coherent.

NO. 1. THE SORITES OF PRO-OBJECTS

Proposition 1. *Let C be a U -category¹ admitting finite colimits. Let h be a functor $C^\circ \rightarrow (\text{Sets})$. The following conditions are equivalent:*

- (i) *h is a filtered colimit, indexed by a small² directed ordered set, of representable functors.*
- (ii) *h is a filtered colimit, indexed by a small filtered category, of representable functors.*
- (iii) *h transforms finite projective limits into finite inductive limits, and there exists a small set of objects such that every element of every $h(X)$ factors through one of them.*

The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) are trivial, and (iii) $\Rightarrow$ (ii) is standard (h is the limit of the h_F indexed by the category of pairs (F, α) with $\alpha \in h(F)$ and F ranging through the full subcategory of C stable under finite projective limits and generated by the given small set). It remains to prove that (ii) $\Rightarrow$ (i). If $\mathcal{L}$ and $\mathcal{M}$ are filtered categories, a functor $F: \mathcal{L} \rightarrow \mathcal{M}$ is said to be cofinal if $\varinjlim_{\mathcal{L}} h_{F(L)}$ is the terminal functor. For every $G: \mathcal{M} \rightarrow \mathcal{N}$ one then has $\varinjlim_{\mathcal{M}} G = \varinjlim_{\mathcal{L}} GF$. Thus it suffices to prove the following.

Lemma 1. *For every small filtered category $\mathcal{L}$, there exists a cofinal functor from a small directed ordered set into $\mathcal{L}$.*

The first projection $\mathcal{L} \times \mathbb{N} \rightarrow \mathcal{L}$ is cofinal ($\mathbb{N}$ being ordered in the usual way); this reduces the problem to the case in which the preorder on $\text{Ob } \mathcal{L}$ defined by $\text{Hom}(X, Y) \neq \emptyset$ has no greatest element. Order by inclusion the set E of finite subcategories of $\mathcal{L}$ having a unique final object (“finite” means that the set of arrows is finite). One gets a functor from E to $\mathcal{L}$ by sending such a subcategory to its final object. Under the above hypothesis, E is filtered and this functor is cofinal.

The functors satisfying the equivalent conditions of Proposition 1 are the Ind-objects of C . They form a full subcategory $\text{Ind } C$ of $\underline{\text{Hom}}(C^\circ, (\text{Sets}))$, stable under colimits. The assignment $C \mapsto \text{Ind } C$ is a functor $(\text{cat}) \rightarrow (\text{cat})$. If filtered colimits exist in C , then for every $h \in \text{Ind } C$ the functor $X \mapsto \text{Hom}(h, h_X)$ is corepresentable; hence there is a functor, denoted $\varinjlim$, from $\text{Ind } C$ to C . In particular one always has a functor $\text{Ind } \text{Ind } C \rightarrow \text{Ind } C$, and every functor $\tilde{C} \rightarrow \text{Ind } D$ extends to a functor $\text{Ind } C \rightarrow \text{Ind } D$. In $\text{Ind } C$, filtered colimits are exact and will again be denoted by

Date: Appendix to *Residues and Duality*, Lect. Notes in Math. 20 (Springer-Verlag, 1966), pp. 404–421
English translation and typeset working edition.

¹ U denotes a fixed universe throughout.

²“small” means “ $\in U$ ”.

“ $\varinjlim$ ”; in particular, identifying C with a full subcategory of $\text{Ind } C$, if $(X_i)_{i \in I}$ is a filtered inductive system in C ,

$$\varinjlim_{i \in I} X_i = \varinjlim_{i \in I} \text{“} \varinjlim \text{”} X_i.$$

For a functor on $\text{Ind } C$ to be representable it is necessary and sufficient that it transform finite projective limits into finite inductive limits, filtered inductive limits into filtered inductive limits, and that its restriction to C satisfy the smallness condition of Proposition 1(iii) (indeed, Proposition 1 then shows that its restriction to C is an Ind -object, to which it is equal everywhere by the condition on limits).

Everything above, except Proposition 1(iii) and the preceding assertion, remains true if one uses throughout limits of sequences (or countable limits, which is the same thing).

By duality one defines the category $\text{pro } C$ of pro-objects (a full subcategory of $\underline{\text{Hom}}(C, (\text{Sets}))^\circ$).

Proposition 2. *Let X be a quasi-compact quasi-separated prescheme (not necessarily Noetherian). The category of quasi-coherent sheaves on X is equivalent to the category of Ind -objects of the category of finitely presented quasi-coherent sheaves on X .*

The functor is “ $\varinjlim$ ” $(\mathcal{F}_i) \mapsto \varinjlim \mathcal{F}_i$. If $\mathcal{F}$ is finitely presented, then for every filtered inductive system $(\mathcal{G}_i)_{i \in I}$,

$$\text{Hom}_X(\mathcal{F}, \varinjlim \mathcal{G}_i) = \varinjlim \text{Hom}_X(\mathcal{F}, \mathcal{G}_i)$$

(by a finite gluing argument commuting with $\varinjlim$, one reduces to the affine case). The preceding functor is therefore fully faithful. Its image is stable under filtered colimits and finite sums. It remains to prove that for every $x \in X$, every sheaf $\mathcal{F}$ on X , and every $s \in \mathcal{F}_x$, there exists a finitely presented sheaf $\mathcal{G}$ and a morphism $\mathcal{G} \rightarrow \mathcal{F}$ whose image contains s :

$\mathcal{F}$ will then be a limit of images of finitely presented sheaves, and each such image a quotient of a finitely presented sheaf by a limit of finite-type subsheaves.

Let $f: \mathcal{C} \rightarrow \mathcal{F}$ be a morphism on some quasi-compact neighborhood U of x , with $\mathcal{C}$ finitely presented and s lying in the image. It suffices to extend $\mathcal{C}$ and f to all of X ; proceeding step by step, it is enough to know how to extend from U to $U \cup V$ when V is affine. This reduces to extending from $U \cap V$ to V :

Lemma 2. *Let $\mathcal{C}$ be a finitely presented sheaf on a quasi-compact open subset U of an affine scheme X , let $\mathcal{F}$ be a sheaf on X , and let f be a morphism from $\mathcal{C}$ to $\mathcal{F}|_U$. Then $\mathcal{C}$ and f can be extended to $\bar{\mathcal{C}}$ and $\bar{f}$ defined on all of X (with $\bar{\mathcal{C}}$ finitely presented).*

Let ι be the inclusion of U into X and let $\mathcal{C}_1$ be the fiber product of $\iota_* \mathcal{C}$ and $\mathcal{F}$ over $\iota_* \iota^* \mathcal{F}$. Then $\mathcal{C}_1$ extends $\mathcal{C}$ and maps to $\mathcal{F}$. If one writes $\mathcal{C}_1$ as a limit of its finite-type subsheaves, then, because U is quasi-compact and $\mathcal{C}$ is finitely presented, one of them already extends $\mathcal{C}$. Writing this subsheaf as a quotient of $\mathcal{O}^n$ by a limit of finite-type subsheaves of $\mathcal{O}^n$, one sees similarly that $\mathcal{C}_1$ can be replaced by a finitely presented sheaf.

Corollary 1. *For a contravariant additive functor $F: (\text{q coh}) \rightarrow \text{Ab}$ to be representable, it is necessary and sufficient that it be left exact and commute with filtered colimits.*

Corollary 2. *Every finitely presented sheaf defined on a quasi-compact open subset U of X extends to a finitely presented sheaf on X .*

Let $\mathcal{A}$ be an abelian category. One will have to work in an essential way inside $\text{pro } D^b \mathcal{A}$, the full subcategory of $\text{pro } D(\mathcal{A})$ consisting of the “ $\varinjlim$ ” of K_i whose cohomology is uniformly bounded with respect to i . The functors $\mathbb{H}^p$ extend to functors $\text{pro } D(\mathcal{A}) \rightarrow \text{pro } \mathcal{A}$.

Proposition 3. *The functors $\mathbb{H}^p$ form a conservative system³ of functors on $\text{pro } D^b \mathcal{A}$.*

³i.e. if u is a morphism such that all the $\mathbb{H}^p(u)$ are invertible, then u itself is invertible.

Let $f: K \rightarrow L$ be a morphism in $\text{pro } D^b \mathcal{A}$ such that the maps $\mathbb{H}^p(K) \rightarrow \mathbb{H}^p(L)$ are isomorphisms (in $\text{pro } \mathcal{A}$). One has to verify that for every M in $D^b(\mathcal{A})$,

$$\text{Hom}(K, M) \leftarrow \text{Hom}(L, M)$$

is an isomorphism. Write $K = \varprojlim K_\alpha$. There is a convergent spectral sequence

$$\text{Ext}^p(\mathbb{H}^q K_\alpha, M) \Longrightarrow \text{Ext}^{p+q}(K_\alpha, M),$$

hence, after passing to the limit, $\varinjlim \text{Ext}^p(\mathbb{H}^q K_\alpha, M) \Longrightarrow \varinjlim \text{Ext}^{p+q}(K_\alpha, M)$, still convergent because q is uniformly bounded. Using the fact that every functor (here Ext) passes to pro-objects, together with the functor $\varinjlim: \text{Ind } \text{Ab} \rightarrow \text{Ab}$, one can rewrite this as

$$\varinjlim \text{Ext}^p(\mathbb{H}^q K, M) \Longrightarrow \varinjlim \text{Ext}^{p+q}(K, M) = \text{Hom}(K[-p-q], M).$$

This spectral sequence remains valid for $M \in D(\mathcal{A})$. The proposition follows immediately from its existence.

NO. 2. FUNDAMENTAL LEMMAS

Recall that the preschemes considered are Noetherian.

Proposition 4 (Duality theorem for an open immersion). *Let U be an open subset of X , let j denote the inclusion, let $\mathcal{J}$ be an ideal defining the complementary closed subset, let $\mathcal{F}$ be a coherent sheaf on U , and let $\mathcal{G}$ be a quasi-coherent sheaf on X . If $\bar{\mathcal{F}}$ is a coherent extension of $\mathcal{F}$, then*

$$(1) \quad \text{Hom}_U(\mathcal{F}, j^* \mathcal{G}) = \text{Hom}_X(\varprojlim \mathcal{J}^n \bar{\mathcal{F}}, \mathcal{G}).$$

The map $\varinjlim \text{Hom}_X(\mathcal{J}^n \bar{\mathcal{F}}, \mathcal{G}) \rightarrow \text{Hom}_U(\mathcal{F}, \mathcal{G})$ is injective: if the image under f of $\mathcal{J}^n \bar{\mathcal{F}}$ in $\mathcal{G}$ has support contained in the complement of U , then it is annihilated by a power of $\mathcal{J}$, say $\mathcal{J}^k$, and the image of $\mathcal{J}^{n+k} \bar{\mathcal{F}}$ is zero.

Assume now that X is affine, and let $f \in \text{Hom}_U(\mathcal{F}, \mathcal{G})$. For k sufficiently large, every section s of $\mathcal{J}^k \bar{\mathcal{F}}$ over X has an image (over U) extending to a section of $\mathcal{G}$ over X . Replacing $\bar{\mathcal{F}}$ by $\mathcal{J}^k \bar{\mathcal{F}}$, one may suppose given a diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & R & \longrightarrow & \mathcal{O}^n & \longrightarrow & \bar{\mathcal{F}} \longrightarrow 0 \\ & & \vdots & & \parallel & & \vdots \\ 0 & \longrightarrow & R_1 & \longrightarrow & \mathcal{O}^n & \longrightarrow & \mathcal{G} \end{array}$$

in which the dotted arrows are defined over U .

For ℓ large enough, by Krull's theorem and the fact that $R/(R \cap R_1)$ is supported outside U , one has $R \cap \mathcal{J}^\ell \cdot \mathcal{O}^n \subset R_1$; this allows one to extend f to $\bar{f}: \mathcal{J}^\ell \bar{\mathcal{F}} \rightarrow \mathcal{G}$.

In the general case, let U_i be a finite affine covering of X and let U_{ijk} be finite affine coverings of the intersections $U_{ij} = U_i \cap U_j$. Since filtered colimits commute with finite products, one gets

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Hom}_U(\mathcal{F}, j^* \mathcal{G}) & \longrightarrow & \prod_i \text{Hom}_{U_i \cap U}(\mathcal{F}, j^* \mathcal{G}) & \rightrightarrows & \prod_{ijk} \text{Hom}_{U_{ijk} \cap U}(\mathcal{F}, j^* \mathcal{G}) \\ & & \uparrow & & \uparrow & & \uparrow \\ 0 & \longrightarrow & \varinjlim \text{Hom}_X(\mathcal{J}^n \bar{\mathcal{F}}, j^* \mathcal{G}) & \longrightarrow & \prod_i \varinjlim \text{Hom}_{U_i}(\mathcal{J}^n \bar{\mathcal{F}}, \mathcal{G}) & \rightrightarrows & \prod_{ijk} \varinjlim \text{Hom}_{U_{ijk}}(\mathcal{J}^n \bar{\mathcal{F}}, \mathcal{G}) \end{array}$$

which completes the proof.

Proposition 4 gives an explicit formula for the functor $(\text{coh on } U) \rightarrow \text{pro}(\text{coh on } X)$ that is left adjoint to j^* . We denote this functor by $j_!$ ("extension by zero")⁴. By Krull's theorem it is exact;

⁴This terminology and notation obviously conflict with the generally accepted ones (Godement's book, SGA 1962, ...). The reader may prefer to read $\mathfrak{J}_!$ instead of $j_!$.

this also follows from the fact that, because X is Noetherian, an injective quasi-coherent sheaf on X remains injective and quasi-coherent after restriction to U .

Proposition 5 (Independence of the compactification).

$$\begin{array}{ccc} U & \hookrightarrow & X \\ \parallel & & \downarrow f \\ U & \hookrightarrow & Y \end{array}$$

Let $f: X \rightarrow Y$ be a proper morphism inducing an isomorphism from the open subset U of Y onto $f^{-1}(U)$. Let $\mathcal{J}$ be a sheaf of ideals defining $Y \setminus U$, and let $\mathcal{I} = f^* \mathcal{J}$. If $\mathcal{F}$ is coherent on X , then

- (i) if $k > 0$, the projective system $\mathbb{R}^k f_* \mathcal{I}^n \mathcal{F}$ is essentially zero (i.e. it defines the zero projective object);
- (ii) if $k = 0$, then for n sufficiently large, $f_* \mathcal{I}^{n+1} \mathcal{F} = \mathcal{J} \cdot f_* \mathcal{I}^n \mathcal{F}$.

One has

$$\bigoplus_{n=0}^{\infty} \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \text{ is a finitely generated } \bigoplus_{n=0}^{\infty} \mathcal{J}^n \text{-module (EGA III),}$$

so for n sufficiently large the map $\mathcal{J} \otimes \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \rightarrow \mathbb{R}^k f_* \mathcal{I}^{n+1} \mathcal{F}$ is surjective; this gives (ii). If $k > 0$, then $\mathbb{R}^k f_*$ vanishes on U ; hence there exists N such that $\mathcal{J}^N \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F}$ is zero for every n . The diagram

$$\begin{array}{ccccc} \mathcal{J}^p \otimes \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} & \longrightarrow & \mathbb{R}^k f_* \mathcal{I}^{n+p} \mathcal{F} & \longrightarrow & 0 \quad (\text{if } n \geq n_0) \\ \downarrow & & \downarrow & & \\ \mathcal{J}^p \cdot \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} & \longrightarrow & \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} & & \end{array}$$

shows that for N sufficiently large and $k > 0$,

$$\mathbb{R}^k f_* \mathcal{I}^{n+N} \mathcal{F} \rightarrow \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \text{ is zero, which is (i).}$$

NO. 3. DEFINITION OF $\mathbb{R}f_!$

A morphism f (respectively a pair of composable morphisms, respectively a triple, and so on) will be called compactifiable if one can respectively find commutative diagrams

$$\begin{array}{ccc} \begin{array}{ccc} X & \hookrightarrow & \overline{X} \\ f \downarrow & \searrow & \\ S & & \end{array} & \begin{array}{ccc} X & \hookrightarrow & \overline{X} \\ f \downarrow & & \downarrow g \\ Y & \hookrightarrow & \overline{Y} \\ h \downarrow & \searrow & \\ S & & \end{array} & \begin{array}{ccc} X & \hookrightarrow & \overline{X} \\ f \downarrow & & \downarrow g \\ Y & \hookrightarrow & \overline{Y} \\ h \downarrow & & \downarrow \\ Z & \hookrightarrow & \overline{Z} \\ \downarrow & \searrow & \\ S & & \end{array} \end{array}$$

in which the horizontal arrows are open immersions and the oblique arrows are proper. Every compactifiable morphism is separated of finite type. Nagata asserts in [2] that his proof shows the converse for Noetherian integral schemes, but the hypotheses he places on the base are not clear.

We wish, for each compactifiable f , to define

$$\mathbb{R}f_! : \text{pro } D_{\text{coh}}^b(X) \rightarrow \text{pro } D_{\text{coh}}^b(S).$$

Recall that $D_{\text{coh}}^b(S)$, the bounded derived category of the category of coherent sheaves on S , is a full subcategory of the derived category of all sheaves (not necessarily quasi-coherent) on S , whose essential image consists of complexes with coherent and bounded cohomology.

If f is proper, one has $\mathbb{R}f_*: D_{\text{coh}}^b(X) \rightarrow D_{\text{coh}}^b(S)$ of finite cohomological dimension, hence an induced functor

$$\mathbb{R}f_!: \text{pro } D_{\text{coh}}^b(X) \rightarrow \text{pro } D_{\text{coh}}^b(S).$$

If f is an open immersion, then

$$f_!: (\text{coh on } X) \rightarrow \text{pro}(\text{coh on } S)$$

induces

$$D^b(\text{coh on } X) \rightarrow D^b \text{pro}(\text{coh on } S),$$

and its image lies in $\text{pro } D_{\text{coh}}^b(S)$; this extends to

$$\mathbb{R}f_!: \text{pro } D_{\text{coh}}^b(X) \rightarrow \text{pro } D_{\text{coh}}^b(S).$$

If the coherent complex K on X is extended to $\bar{K}$ on S , its image is “ $\varprojlim$ ” $\mathcal{J}^n \bar{K}$, where $\mathcal{J}$ defines $S \setminus X$.

For a composite $f = gh$ with g proper and h an open immersion, define $\mathbb{R}f_! = \mathbb{R}g_! \mathbb{R}h_!$. One has to verify:

Independence of the compactification.

Consider a commutative diagram

$$\begin{array}{ccc} & & X' \\ & \nearrow j' & \downarrow g \\ X & & \\ & \searrow j'' & \downarrow \\ & & X'' \\ f \downarrow & \nwarrow \bar{f} & \\ S & & \end{array}$$

where j' and j'' are open immersions and $\bar{f}$ and g are proper maps. One has $j'X = g^{-1}j''X \cap j'X$. One must prove that

$$\mathbb{R}\bar{f}_* \mathbb{R}j_!'' = \mathbb{R}(\bar{f}g)_* \mathbb{R}j_!'$$

or, since $\mathbb{R}(\bar{f}g)_* = \mathbb{R}\bar{f}_* \mathbb{R}g_*$, equivalently that

$$\mathbb{R}j_!'' = \mathbb{R}g_* \mathbb{R}j_!'$$

One may reduce to the case where X is dense in X' , so as to be in the situation of Proposition 5. Let K be a bounded coherent complex on X , extended to $\bar{K}$ on X' . Then $g_* \bar{K}$ extends K on X'' , and there is a morphism

$$(2) \quad g_* \mathcal{J}'^n \bar{K} \rightarrow \mathbb{R}g_* \mathcal{J}''^n \bar{K},$$

where $\mathcal{J}'$ denotes an ideal sheaf defining $X' \setminus X$. Proposition 5(ii) shows that

$$j_!'' K = \text{“}\varprojlim\text{” } g_* \mathcal{J}''^n \bar{K}.$$

Passing to the “limit” one gets

$$\text{“}\varprojlim\text{” } \mathbb{R}^p g_* \mathcal{J}''^n \bar{K}^q \implies \text{“}\varprojlim\text{” } \mathbb{R}^{p+q} g_* \mathcal{J}''^n \bar{K};$$

Proposition 5(i) shows that this spectral sequence degenerates, and it follows from Proposition 3 that the morphism induced from (1) by passage to the “limit” is an isomorphism. This is exactly the desired formula.

It remains to verify

- (a) for every diagram, a compatibility

$$\begin{array}{ccc} X & \longrightarrow & X' \\ f \downarrow & & \downarrow h \\ S & \longrightarrow & \bar{X} \end{array} \quad \mathbb{R}\bar{f}_* \mathbb{R}j'_! = \mathbb{R}\bar{f}_* \mathbb{R}g_* \mathbb{R}j'_! = \mathbb{R}\bar{f}_* \mathbb{R}g_* \mathbb{R}h_* \mathbb{R}j'_! = \mathbb{R}\bar{f}_* \mathbb{R}(gh)_* \mathbb{R}j'_! = \mathbb{R}f_* \mathbb{R}j'_!;$$

- (b) for every compactifiable pair, an identification

$$\mathbb{R}(fg)_! = \mathbb{R}f_! \mathbb{R}g_!;$$

- (c) for every compactifiable triple, a compatibility

$$\begin{array}{ccc} \mathbb{R}(fgh)_! & = & \mathbb{R}(fg)_! \mathbb{R}h_! \\ \parallel & & \parallel \\ \mathbb{R}f_! \mathbb{R}(gh)_! & = & \mathbb{R}f_! \mathbb{R}g_! \mathbb{R}h_! \end{array}$$

One then obtains:

Theorem 1. *For a compactifiable morphism $f: X \rightarrow Y$, for $K \in \text{pro } D_{\text{coh}}^b(X)$ and $L \in \text{pro } D_{\text{coh}}^b(Y)$, let*

$$\text{Hom}_f(K, L) = \text{Hom}_Y(\mathbb{R}f_! K, L).$$

Modulo compactifiability questions, this makes the categories $\text{pro } D_{\text{coh}}^b(X)$ into a cofibration over the category of Noetherian preschemes (with compactifiable morphisms as arrows, and so on).

NO. 4. DEFINITION OF $\mathbb{R}f^!$

It follows from Verdier's general formalism [1] that the functor $\mathbb{R}f_*: D(X) \rightarrow D(S)$ has a right adjoint $D^+(S) \rightarrow D^+(X)$. It deserves the name $\mathbb{R}f^!$ only when f is proper, and its definition is not immediately suited to computation.

One must first make explicit a procedure for computing $\mathbb{R}f_*$ at the level of complexes, of the form $\mathbb{R}f_* K = f_* C^*(K)$, where C^* is a finite acyclic resolution depending functorially and exactly on the object to which it is applied and compatible with filtered colimits. Then, for every injective I on S , the functor $\text{Hom}(f_* C^p \mathcal{F}, I)$ is exact in quasi-coherent sheaves on X and transforms $\varinjlim$ into $\varprojlim$, hence is representable by some $f_p^! I$, which is injective and quasi-coherent. The $f_p^! I$ form a complex, and this defines

$$\mathbb{R}f^!: D^+ S \rightarrow D^+ X,$$

right adjoint to

$$\mathbb{R}f_*: D(X) \rightarrow D(S).$$

Here is how one may define such a resolution.

- (1) If S is separated, choose a finite covering of X by open subsets affine over S (for example affine open subsets) and put

$$\mathbb{R}f_* K = f_* \check{C}(\mathcal{U}, K)$$

(the alternating Čech complex).

- (2) Otherwise, the canonical flasque resolutions have here two defects.

- (a) $f_* C^* \mathcal{F}$ is no longer quasi-coherent. This does not matter as long as, as here, injective quasi-coherent sheaves are injective when regarded simply as sheaves.
- (b) C^* does not commute with filtered colimits. It is easy to correct this defect: write a quasi-coherent sheaf as $\mathcal{F} = \varinjlim \mathcal{F}_i$ with the $\mathcal{F}_i$ finitely presented, and set

$$C'^* \mathcal{F} = \varinjlim C^* \mathcal{F}_i.$$

This is independent of the choices made.

For an open immersion one takes $\mathbb{R}f^!K = f^*K$.

For a composite morphism $f = gh$ with g proper and h an open immersion, one takes $\mathbb{R}f^! = \mathbb{R}g^!\mathbb{R}h^!$. Putting together Proposition 4 and the preceding discussion one obtains:

Theorem 2 (Duality). *Let $f: X \rightarrow S$ be compactifiable, with $f = gh$. The functors*

$$\mathbb{R}f_!: \text{pro } D_{\text{coh}}^b(X) \rightarrow \text{pro } D_{\text{coh}}^b(S)$$

and

$$\mathbb{R}f^!: D^+(S) \rightarrow D^+(X)$$

are adjoint to one another.

One should still verify the independence of the compactification and a list of compatibilities and identifications analogous to those for $\mathbb{R}f_!$. Some special cases of the formalism follow immediately from the adjunction formula (uniqueness of an actual adjoint, and so on).

In the general case one may rely on the following proposition.

Proposition 6. *Let $f: X \rightarrow S$ be compactifiable and let $L \in D_{\text{qc}}^+(S)$. The stalk at $x \in X$ of $\mathcal{H}^p\mathbb{R}f^!L$ is given by*

$$(\mathcal{H}^p\mathbb{R}f^!L)_x = \varinjlim_{x \in U} \text{Hom}_S^p(\mathbb{R}f_!(U, \mathcal{O}_U), L).$$

One may assume X proper and L given as a complex of injective quasi-coherent sheaves. Let U be an affine open subset of X and let $\mathcal{J}$ define $X \setminus U$. Then

$$\begin{aligned} (\mathcal{H}^p\mathbb{R}f^!L)(U) &= \mathbb{H}^p(\mathbb{R}f^!L(U)) = \mathbb{H}^p \text{Hom}_U(\mathcal{O}, \mathbb{R}f^!L) \\ &= \mathbb{H}^p \varinjlim \text{Hom}_X(\mathcal{J}^n, \mathbb{R}f^!L) = \varinjlim \text{Hom}_S^p(\mathbb{R}f_*\mathcal{J}^n, L) \\ &= \text{Hom}_S^p(\mathbb{R}f_!(U, \mathcal{O}_U), L), \end{aligned}$$

and the proposition follows.

We shall henceforth assume for $\mathbb{R}f^!$ a formalism of variance analogous to that of Theorem 1 for $\mathbb{R}f_!$.

If one wants to make the definition of $\mathbb{R}f^!$ in the proper case more explicit, one may remark that if a quasi-coherent sheaf $\mathcal{F}$ represents a functor F , then for every open subset U and every ideal $\mathcal{J}$ defining $X \setminus U$ one has

$$\mathcal{F}(U) = \varinjlim \text{Hom}(\mathcal{J}^n, \mathcal{F}) = \varinjlim F(\mathcal{J}^n).$$

NO. 5. COMPUTATION OF $\mathbb{R}f^!$

For a finite morphism, $\mathbb{R}f^!$ is what one expects; to see this it suffices to take, in the computation of $\mathbb{R}f_*$, the identity as the resolving functor. For projective space, uniqueness of an adjoint leaves no choice (one uses the fact that the adjunction formula holds for $\mathbb{R}f_! = \mathbb{R}f_*: D(X) \rightarrow D(S)$); we return to this morphism. For an open immersion, $\mathbb{R}f^! = \mathbb{R}f^*$.

For every quasi-projective open subset U of X over S , one can therefore compute the restriction of $\mathbb{R}f^!$ to U . In particular, if one can verify locally that $\mathbb{R}^q f^!K = 0$ for $q \neq p$, then one knows $\mathbb{R}f^!K$, because an object of the derived category having only one nonzero cohomology sheaf is determined by that sheaf. Let us recall an important case in which this condition is satisfied, with $K = \mathcal{O}$.

Proposition 7. *Let $f: X \rightarrow Y$ be compactifiable. Assume that locally (on X and on Y) one can find a diagram*

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow g & \swarrow h \\ & S & \end{array}$$

in which g and h are local complete intersections of relative dimensions d' and d'' , and put $d = d' - d''$.

Then $\mathbb{R}f^!\mathcal{O}$ has only one nonzero cohomology sheaf, located in degree $-d$, and that sheaf is invertible.

Indeed, $\mathbb{R}g^!\mathcal{O} = \mathbb{R}f^!\mathbb{R}h^!\mathcal{O}$. One knows that both $\mathbb{R}g^!\mathcal{O}$ and $\mathbb{R}h^!\mathcal{O}$ are of the indicated form, concentrated in degrees $-d'$ and $-d''$, by local computations in projective space. Since everything is local, one may suppose $\mathbb{R}^{-d''}h^!\mathcal{O} \cong \mathcal{O}$, and the assertion follows.

To pass from this to more general complexes one would first need formulas of the shape

$$\mathbb{R}f^!(K \overset{\mathbb{L}}{\otimes} L) = \mathbb{R}f^!(K) \overset{\mathbb{L}}{\otimes} \mathbf{L}f^*L.$$

Only the definition of the morphism presents any difficulty, because its being an isomorphism is a local problem. Degree restrictions are of course needed. I have not checked in detail any of what follows.

First method for defining the morphism.

Assume K and L lie in $D_{\text{coh}}^b(S)$ (and are not merely quasi-coherent), and moreover that

- (a) $\mathbb{R}f^!K$ is bounded in degree (it is automatically coherent; that is a local problem),
- (b) $K \overset{\mathbb{L}}{\otimes} L$ is bounded in degree,
- (c) $\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbf{L}f^*L$ is bounded in degree.

To define

$$\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbf{L}f^*L \longrightarrow \mathbb{R}f^!(K \overset{\mathbb{L}}{\otimes} L),$$

it suffices by duality to define

$$\mathbb{R}f_!(\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbf{L}f^*L) \longrightarrow K \overset{\mathbb{L}}{\otimes} L.$$

Now by K"unneth,

$$\mathbb{R}f_!(\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbf{L}f^*L) = \mathbb{R}f_!\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} L,$$

and the desired map is induced by the adjunction morphism $\mathbb{R}f_!\mathbb{R}f^!K \rightarrow K$.

Second method.

Assume f compactifiable, flat, and relatively local complete intersection. Then $\mathbb{R}f^!\mathcal{O} = \omega[d]$ with ω invertible. Let $\mathbb{R}f_!$ denote some complex-level procedure computing $\mathbb{R}f_!$ (that is, such a procedure for a compactification $\bar{f}: \bar{X} \rightarrow S$ of f), assumed to be of the sort described in No. 4 and such that $C^p\mathcal{F}$ vanishes for $p > d$ (which can be achieved by truncation). There is then a morphism

$$\mathbb{R}^d\bar{f}_*\mathcal{F}[-d] \rightarrow \mathbb{R}\bar{f}_*\mathcal{F} \quad \text{for every sheaf } \mathcal{F} \text{ on } \bar{X}$$

(the last cohomology sheaf maps to its complex). Hence for any injective I on S one gets morphisms of complexes

$$\text{Hom}_S(\mathbb{R}\bar{f}_*\mathcal{F}, I) \rightarrow \text{Hom}_S(\mathbb{R}^d\bar{f}_*\mathcal{F}[-d], I).$$

Since $\mathbb{R}^d\bar{f}_*[-d]$ is right exact in $\mathcal{F}$, the functor in $\mathcal{F}$ on the right is representable: define $I_!$ by

$$\text{Hom}_S(\mathbb{R}^d\bar{f}_*\mathcal{F}, I) = \text{Hom}_{\bar{X}}(\mathcal{F}, I_!) \quad \text{and} \quad \mathbb{R}f_!I = I_![d].$$

The preceding construction defines

$$\mathbb{R}f^!I \rightarrow \mathbb{R}f_!^!I,$$

and for a complex of injectives the functor $\mathbb{R}f_!^!$ is defined termwise.

It remains to evaluate $\mathbb{R}f_!^!I$ on affine open subsets U of X ; this is given by $\varinjlim \text{Hom}(\mathbb{R}^d f_!\mathcal{J}^n, I)$. Since this problem is local upstairs, it is easy to show by projective methods that the dual of " $\varinjlim$ " $\mathbb{R}^d f_!\mathcal{J}^n = \mathbb{R}^d f_!(U, \mathcal{O})$, which is an ind-object, is $f_*(U, \omega)$, represented as the limit of its submodules of sections having poles of order at least n outside U ($n \rightarrow \infty$).

This gives $\mathbb{R}f_!^* I = \omega \otimes f^* I[d]$, and hence the general formula

$$\mathbb{R}f^! K = f^* K \otimes \omega[d].$$

BIBLIOGRAPHY

- [1] J.L. Verdier, Séminaire Bourbaki, November 1965, exposé 300.
- [2] M. Nagata, *Imbedding of an abstract variety in a complete variety*, Journal of Mathematics of Kyoto University, vol. 2, no. 1, 1962.